

# Xiaoqing PAN

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NUCLEATE Office, Julius-Maximilians-Universität Würzburg

Re: Postdoctoral Researchers

Dear Sir or Madam,

My name is Xiaoqing, I just finished my PhD study at Hans-Knöll-Institute in Jena, Germany. I am writing to apply for the open postdoc position at the University of Wuerzburg.

I studied microbiology. My PhD project was focusing on double-stranded RNA metabolism in both humans and pathogenic fungi. On one hand, I was investigating host ADAR1-mediated A-to-I RNA editing pattern changes during fungal stimulations. Through characterizing transcriptomic responses after fungal challenges with a focus on host alternative splicing and A-to-I RNA editing, we surprisingly found out that *Rhizopus oryzae*, a human fungal pathogen that causes mucormycosis, can induce A-to-I hyper-editing as well as disrupt host RNA splicing. I contributed to analyse multiple RNA-seq cohorts, DEG analysis via DESeq2, alternative splicing analysis via rMATS-Turbo, RNA structure analysis via SHAPE-Map and RNA editing analysis. Work has resulted in one review and one draft in preparation. On the other hand, I was elucidating how the opportunistic pathogen *Aspergillus fumigatus* produces small RNAs, how dicer-like proteins process dsRNA substrates, and how excess dsRNA arrest fungal growth. I also combined bioinformatics (sRNA-seq and mRNA-seq analyses) and experimental validations (qPCR as well as Northern blot) to characterize the small RNA landscape and the contribution of RNAi machinery to fungal biology. This work has resulted in two publications on *RNA* and *Molecular microbiology*. Through my PhD training, I've been equipped with solid wet-lab skills, such as fungi infecting human cell assay, microbial culturing, DNA/RNA extraction, and gained experiences on bioinformatics.

I would like to develop an academic career in RNA research. I believe that postdoctoral training is essential for my career development. And I find my background matches **Prof. Dr. Alexander Westermann** and **Prof. Dr. Julian König** groups. I was interested in sRNA functions in fungi and would be interested in expanding its functional exploration in bacteria. I was also analyzing sRNA-seq and mRNA-seq in both fungi and human hosts. Correspondingly, Prof. Westermann's research interest is advanced cross-species RNA-seq approaches and sRNA functions in bacteria. RNA editing, as a type of RNA modification, together with RNA splicing are two important post-transcriptional regulators. While I was illustrating the RNA editing and RNA splicing patterns change during fungal challenges, Prof. König focuses on investigating mechanisms of splicing regulation in human disease and RNA modifications in gene expression and dosage compensation.

I studied RNA regulation in fungal biology, and I'd like to take the opportunity to complement my skillset on other microorganisms, such as bacteria. This is realistic since RNA-seq analysis and RNA detection assays are globally used. Second, I began to investigate host response in my PhD, which created my interest in better understanding human RNA regulation and interferon response, I believe this position is an opportunity to step forward to it. Third, I am seeking an international collaborative work environment, as I find integration into a vibrant team important for my development and performing high-quality science. Julius-Maximilians-Universität is a very international university in Germany and famous for RNA research in bacteria. I believe that bringing my perspective from fungal genetics and host-pathogen immunology could contribute fresh insights to related projects in the group.

Thank you for your time and consideration.

Best,

Xiaoqing

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## SUMMARY

RNA biologist and computational transcriptomics researcher with expertise in high-throughput sequencing, gene regulation, and RNA-mediated cellular responses. Experienced in integrating molecular biology with bioinformatics to study small RNAs, RNA interference, alternative splicing, and RNA editing across host and microbial systems. Proficient in RNA-seq analysis, reproducible computational workflows (R, Bash), and sequencing-based approaches for investigating post-transcriptional regulation.

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## TECHNICAL SKILLS

- **Molecular Biology:** sRNA cloning, NGS library preparation, GUS reporter assays, luciferase and GFP reporter assays, Gibson Assembly Cloning, Genetic Engineering, Northern Blot, qPCR, PCR, immunostaining, LC-MS, cultivation of microorganisms (BSL-1/2), plant cultivation and tissue preparation
- **Bioinformatics:** RNA-seq, sRNA-seq, differential expression, alternative splicing, small RNA characterization and target prediction, data visualization, HPC (Slurm), UCSC, SwissProt, KEGG, NCBI
- **Programming:** R, Python, Bash, Git, Docker, Linux
- **Language:** Chinese (native), English (fluent), German (B1)

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## EXPERIENCES

### RNA landscape of human fungal pathogen *Aspergillus fumigatus*

- Characterized RNAi and tRNA fragments processing pathways in *A. fumigatus* across developmental stages (conidia, mycelium), revealing morphotype-specific RNA regulatory mechanisms
- Analyzed differential expressed genes on mRNA-, sRNA-seq datasets, uncovering post-transcriptional regulatory networks controlling fungal development and stress response.
- Validated genes of interest (sRNA, tRNA fragments, sRNA targets) via qPCR, northern blot

### Human PBMC transcriptome changes upon fungal stimulations

- Analyzed pathogen-specific transcriptomic signatures in human immune cells responding to fungal pathogens, characterized differential expressed genes, alternative splicing, A-to-I RNA editing, pathogen recognition receptor pathway activation
- Built HPC pipeline, wrote custom R scripts for multi-condition cohort analysis

### Human neutrophils exporting extracellular RNA upon *A. fumigatus* confrontation

- Analyzed EV-associated sRNA, tRNA, and mRNA cargo from primary human neutrophils during fungal challenge

### Non-coding RNA analysis of rice-*Xanthomonas oryzae* interaction

- Studied how T3SS-dependent bacterial effectors from *Xanthomonas oryzae* shape host non-coding RNA regulation; performed GUS reporter assays, plant cultivation, tissue preparation, LC-MS, qPCR, and PCR
- Engineered overexpressed and knockout mutants of circular RNA and microRNA in rice, established genetic tools that enabled investigation of ncRNA-mediated regulation

## Teaching

- Microbiology and Molecular Biology, MMB005, 2022-2025. Supervised master students investigating gene regulation together with Dr. Blango and other colleagues.

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## EDUCATION

**PhD** | Microbiology, Friedrich Schiller University, Germany (supervisor: Dr. Matthew Blango) 2026

*Thesis: RNA regulation in host-fungal interactions: investigation of the host epitranscriptome and fungal small RNAs*

**MSc** | Bioengineering, Chinese Academy of Sciences, China (supervisor: Dr. Kuaifei Xia) 2021

*Thesis: Non-coding RNAs ciR133 and miR168a-5p play roles in response to Xoo and abiotic stress in rice, respectively*

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## PUBLICATIONS

- **Pan X\***, Kelani AA\*, Schrettenbrunner L, Prabakar S, Israni B, Blango MG. (2026). An improved description of the small RNA landscape of the human fungal pathogen *Aspergillus fumigatus*. Mol Microbiol 1-12.
- **Pan X\***, Bruch A\*, Blango MG. (2024). The past, present, and future of RNA modifications in infectious disease research. ACS Infect Dis 10(12), 4017-4029.
- Xia K\*, **Pan X\***, Chen H, Xu X, Zhang M. (2023). X00-responsive transcriptome reveals the role of the circular RNA133 in disease resistance by regulating expression of OsARAB in rice. Phytopathol Res 5, 22.
- Xia K\*, **Pan X\***, Zeng X, Zhang M. (2023). Rice miR168a-5p regulates seed length, nitrogen allocation and salt tolerance by targeting OsOFP3, OsNPF2.4, and OsAGO1a, respectively. J. Plant Physiol 208, 153905.
- Kelani AA, Bruch A, Riviaccio F, Visser C, Krüger T, Weaver D, **Pan X**, Schaeuble S, Panagiotou G, Kniemeyer O, Bromley MJ, Bowyer P, Barber AE, Brakhage AA, Blango MG. (2023). Disruption of the *A. fumigatus* RNA interference machinery alters the conidial transcriptome. RNA 29(7), 1033-1050.

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## CONFERENCES

- **2025 - Oral presentation:** German speaking Mycological Society (DMykG), Germany
- **2025 - Poster:** Gordon Research Seminar - RNA Editing, Italy
- **2024 - Oral presentation:** Jena RNA Club Symposium, Germany
- **2024 - Poster:** RNA Society Meeting, UK
- **2023 - Poster:** EMBL - The Non-coding Genome Symposium, Germany

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## AWARDS & SCHOLARSHIP

- DAAD Scholarship 2021 - 2025
- Jena School for Microbial Communication (JSMC) travel grant 2023

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## REFEREE

- Dr. Matthew Blango ([matthew.blango@leibniz-hki.de](mailto:matthew.blango@leibniz-hki.de))
- Dr. Olaf Kniemeyer ([olaf.kniemeyer@leibniz-hki.de](mailto:olaf.kniemeyer@leibniz-hki.de))